

MATH 131P WINTER 2025 FINAL PROJECT

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The Hanging Chain: Exploring the Mathematics of Transverse Oscillations

1. INTRODUCTION

The study of Partial Differential Equations (PDEs) provide a basis for understanding the dynamics of many physical systems. Additionally, by applying boundary value conditions (so that we may arrive at a Boundary Value Problem (BVP)), we may observe these systems in various states. We seek to describe the motion of a chain of length L , hanging downwards, where one end is fastened in place at $x = L$. We define the x – *axis* to be pointing upwards and the u – *axis* to be pointing right. The goal of this paper is to describe the transverse oscillations of this chain via a boundary value problem. We solve for $u(x, t)$, a function that describes the wavelike motions, perpendicular to the chain's length, that the chain undergoes when displaced. We begin by deriving the PDE for this chain's movement. Next, we establish the BVP that we can apply to this PDE. Finally, we solve the BVP and conduct an exploration of the physical consequences of our results. This paper will show that the motion of a hanging chain, as described above, is dependent on both time and position along the chain.

2. PARTIAL DIFFERENTIAL EQUATION DERIVATION

To derive the PDE that represents the transverse oscillations of a chain, we begin by defining the properties of this chain. We make two assumptions about our chain: (1) we have a homogeneous, purely elastic chain and (2) the motion of this chain is transverse only. A homogeneous chain is a chain whose material is uniform throughout its entire length. It follows that this chain has constant mass density (ρ), tension (τ), and elasticity for all points along its length. Additionally, a chain that offers no resistance to bending and returns to its original shape once a displacing force is removed is defined as a purely elastic chain. Finally transverse motion describes the movement of the chain perpendicular to its length. This means we can assume the chain does not move along its length (along the x – *axis*) but rather back-and-forth along the u – *axis*. Additionally we assume these transverse vibrations are small and that the slope along the chain, $\frac{\partial u}{\partial x}(x, t)$, at all points is also small.

We note that since the tension in the chain is constant across the entire chain, we can find the tension at equilibrium (no motion) and assume that this remains constant with motion. Thus we see, given that density ρ is equal to $\frac{m}{l}$, the tension at any point in the chain is

$$\begin{aligned}\tau(x) &= W + m * a \\ &= m_x * g + m * (0) \\ &= m \frac{x}{l} * g + 0 \\ &= \rho g x.\end{aligned}\tag{1}$$

The tension of this chain is tangent to the motion of the chain at any point. This is due to the fact that the chain is purely homogeneous. Now we consider two points on the chain that are separated by a distance Δx . Since we have only transverse motion we can apply Newton's Second Law $f = ma$ to the horizontal direction of the chain's motion, where the only forces acting on this chain are the tension forces at the two endpoints. We note that the mass of chain between these two endpoints is equal to $\Delta x * \rho$. Additionally the acceleration of the chain is $\frac{\partial^2 u}{\partial t^2}$.

Plugging this in to Newton's Second Law, we see that:

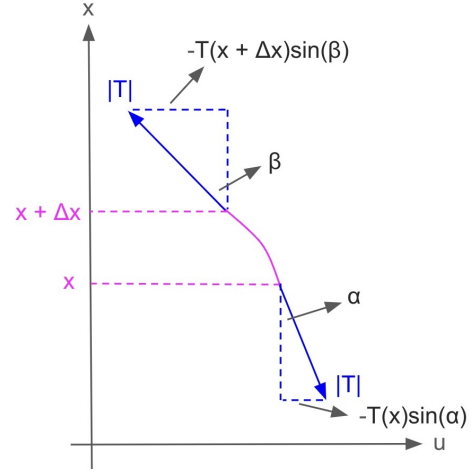
$$F = m * a$$

$$\tau(x) * \sin(\alpha) - \tau(x + \Delta x) * \sin(\beta) = p\Delta x * \frac{\partial^2 u}{\partial t^2} \quad (2)$$

[1, Equation 3.2.1].

Example 2.1. Diagram of Tension in Chain

Equation (2) uses the fact that, for any two points on the chain, the magnitude of the tensions at those points is the same. However, the tension forces at these two points are pointed in opposite horizontal directions. This fact is demonstrated to the right.



Now we use the fact that for small angles $\cos(x) \approx 1$ and, accordingly, $\sin(x) \approx \tan(x)$. We then apply the fact that the tangent values at these points are equivalent to the slopes of u at these points. Plugging this into (2) we get:

$$\tau(x) * \tan(\alpha) - \tau(x + \Delta x) * \tan(\beta) = p\Delta x * \frac{\partial^2 u}{\partial t^2} \quad (3)$$

$$\frac{1}{\Delta x}(\tau(x) * \frac{\partial u}{\partial x}(x, t) - \tau(x + \Delta x) * \frac{\partial u}{\partial x}(x + \Delta x, t)) = p * \frac{\partial^2 u}{\partial t^2}$$

[1, Equation 3.2.2].

As we let $x \rightarrow 0$ and plug in our value for tension (1) we see that:

$$\frac{\partial}{\partial x}(pgx * \frac{\partial u}{\partial x}) = p * \frac{\partial^2 u}{\partial t^2}$$

$$g * \frac{\partial}{\partial x}(x * \frac{\partial u}{\partial x}) = \frac{\partial^2 u}{\partial t^2} \quad (4)$$

$$g * (x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x}) = \frac{\partial^2 u}{\partial t^2}.$$

Thus we arrive at our final PDE which represents the transverse oscillations of our chain.

Proposition 2.2. PDE for Hanging Chain [1, Equation 3.2.6.c]

$$\frac{\partial^2 u}{\partial t^2} = g * (x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x})$$

3. BOUNDARY VALUE PROBLEM LAYOUT

We now establish the BVP that we can apply to the PDE given by Proposition 2.2. Given that we are interested in observing the motion of this chain, we look at the values of x where $0 \leq x \leq L$. Then, since we hold one end of the chain fixed, we see that $u(L, t) = 0$. Additionally we are interested in observing this chain after it is displaced and set in motion so we only observe this chain for time $t \geq 0$.

Now we have two functions of x that describe the initial state of this chain. Firstly, we define $u(x, 0) = f(x)$ to be the initial displacement. Additionally, we define $\frac{\partial u}{\partial t}(x, 0) = v(x)$ to be the initial velocity. Finally, we define $u(0, t)$ to be finite. That is, we physically constrain the vibrations of the free end of the chain to be bounded. This helps us later to rule out unbounded solutions to u that do not have physical interpretations.

Proposition 3.1. *BVP for Hanging Chain (using PDE given in Proposition 2.2) [1, Proposition 6.3.1]*

$$\begin{aligned} 0 &\leq x \leq L, \quad t \geq 0 \\ u(0, t) &\text{ is finite, } \quad u(L, t) = 0 \\ u(x, 0) &= f(x), \quad \frac{\partial u}{\partial t}(x, 0) = v(x) \\ \frac{\partial^2 u}{\partial t^2} &= g \left(x \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} \right) \end{aligned}$$

4. BOUNDARY VALUE PROBLEM SOLUTION

We now solve the above boundary value problem, Proposition 3.1, by assuming that we can apply the Theorem of Separation of Variables.

Theorem 4.1. *Separation of Variables*

If we have a linear, homogeneous PDE then its solutions can be expressed as a product of functions, each depending on only one variable. Assuming this product form for the solution allows for the separation of the function into different components, each depending on one variable. This allows for integration of each variable separately.

We assume the solution has the separated form $u(x, t) = X(x)T(t)$. Plugging this into Proposition 2.2 we find the following:

$$\begin{aligned} X(x)T''(t) &= g(x * X''(x)T(t) + X'(x)T(t)) \\ \frac{1}{g} * \frac{T''(t)}{T(t)} &= \frac{x * X''(x)}{X(x)} + \frac{X'(x)}{X(x)}. \end{aligned} \tag{5}$$

Now we note that the left side of (5) is a function of only t and likewise the right side of (5) is a function of only x . Thus we have achieved the situation described by Theorem 4.1. Accordingly, since both of these sides must equal the same value, independent of each other, we set them both equal to the same constant value λ so that we have:

$$\frac{1}{g} * \frac{T''(t)}{T(t)} = \lambda \quad \text{and} \quad \frac{x * X''(x)}{X(x)} + \frac{X'(x)}{X(x)} = \lambda.$$

This gives us two separated equations to solve for, which we can later plug into $u(x, t) = X(x)T(t)$ to achieve a final solution. We observe the following two equations for this BVP.

Proposition 4.2. *Separated Equation for Time [1, Equation 6.3.2]*

$$T''(t) - \lambda g T(t) = 0$$

Proposition 4.3. *Separated Equation for Horizontal Displacement [1, Equation 6.3.3]*

$$xX''(x) + X'(x) - \lambda X(x) = 0$$

We now seek to solve these two separated equations, beginning with Proposition 4.2.

Proposition 4.4. *Solutions to Proposition 4.2*

The roots of the characteristic polynomial for this equation give us

$$r^2 - g\lambda = 0 \Rightarrow r^2 = g\lambda \Rightarrow r = \pm\sqrt{\lambda g}.$$

The general solution is given by

$$T(t) = \begin{cases} Ae^{\sqrt{\lambda g}t} + Be^{-\sqrt{\lambda g}t}, & \text{if } \lambda > 0 \\ A + Bt, & \text{if } \lambda = 0 \\ A\cos(\sqrt{|\lambda|g}t) + B\sin(\sqrt{|\lambda|g}t), & \text{if } \lambda < 0 \end{cases}$$

for scalars $A, B \in \mathbf{R}$.

We first observe that if $\lambda \geq 0$ we arrive at solutions that are either exponential or linear. We should ignore these cases because we do not expect the motion of the chain to grow linearly or exponentially with time. Thus we look only at cases where, in both Proposition 4.2 and 4.3 (since λ is constant between them), $\lambda < 0$. The solution to Proposition 4.2 is:

$$T(t) = A\cos(\sqrt{|\lambda|g}t) + B\sin(\sqrt{|\lambda|g}t). \quad (6)$$

[1, Equation 6.3.4]. Now we look to solve for Proposition 4.3. We consolidate the first two terms to find

$$(xX'(x))' - \lambda X(x) = 0 \quad (7)$$

which is a Sturm-Liouville Problem (SLP).

Definition 4.5. Singular Sturm-Liouville Problem

A SLP is a way to reduce a second-order, homogeneous, linear ODE over a finite interval $[a, b]$. It has the form:

$$(p(x)y')' + (q(x) + \lambda w(x))y = 0, \quad a < x < b$$

where $p(x)$, $q(x)$, and $w(x)$ (the weight function) are coefficient functions and $y(x)$ is an unknown function. A SLP is regular if $p(x)$, $p'(x)$, $q(x)$, and $w(x)$ are continuous on the interval $[a, b]$, $p(x) > 0$, and $w(x) > 0$. If any of these conditions fail or if the BVP is defined over an infinite interval, the SLP is singular.

We have $p(x) = x$, $q(x) = 0$, and $w(x) = 1$. Since $p(0) = 0$, this is a singular SLP.

To transform this SLP into a form that we can solve, we perform the following change of variables: $a = 2\sqrt{x}$. We begin by assuming that $X(x) = A(a)$. Now we seek to find $\frac{\partial X}{\partial x}$ and $\frac{\partial^2 X}{\partial x^2}$ in terms of a :

$$\frac{\partial X}{\partial x} = \frac{\partial X}{\partial a} * \frac{\partial a}{\partial x} = \frac{\partial X}{\partial a} * \frac{1}{\sqrt{x}} \quad (8)$$

$$\begin{aligned} \frac{\partial^2 X}{\partial^2 x} &= \frac{\partial}{\partial x} \left(\frac{\partial X}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{\partial X}{\partial a} * \frac{1}{\sqrt{x}} \right) = \frac{\partial}{\partial x} \left(\frac{\partial X}{\partial a} \right) * \frac{1}{\sqrt{x}} + \frac{\partial X}{\partial a} * \frac{\partial}{\partial x} \left(\frac{1}{\sqrt{x}} \right) \\ &= \frac{\partial}{\partial a} \left(\frac{\partial X}{\partial x} \right) * \frac{1}{\sqrt{x}} - \left(\frac{1}{2x^{3/2}} \right) * \frac{\partial X}{\partial a} \\ &= \frac{\partial}{\partial a} \left(\frac{\partial X}{\partial a} * \frac{1}{\sqrt{x}} \right) * \frac{1}{\sqrt{x}} - \left(\frac{1}{2x^{3/2}} \right) * \frac{\partial X}{\partial a} \\ &= \frac{1}{x} * \frac{\partial^2 X}{\partial a^2} - \left(\frac{1}{2x^{3/2}} \right) * \frac{\partial X}{\partial a}. \end{aligned} \quad (9)$$

Now, noting that $x = \frac{a^2}{4}$, we plug the above results into Proposition 4.3 to find:

$$\begin{aligned} x \left(\frac{1}{x} * \frac{\partial^2 X}{\partial a^2} - \left(\frac{1}{2x^{3/2}} \right) * \frac{\partial X}{\partial a} \right) + \frac{\partial X}{\partial a} * \frac{1}{\sqrt{x}} - \lambda X &= 0 \\ \frac{a^2}{4} \left(\frac{4}{a^2} * \frac{\partial^2 X}{\partial a^2} - \left(\frac{8}{2a^3} \right) * \frac{\partial X}{\partial a} \right) + \frac{\partial X}{\partial a} * \frac{2}{a} - \lambda X &= 0 \\ \frac{\partial^2 X}{\partial a^2} - \frac{1}{a} * \frac{\partial X}{\partial a} + \frac{\partial X}{\partial a} * \frac{2}{a} - \lambda X &= 0 \\ \frac{\partial^2 X}{\partial a^2} + \frac{1}{a} * \frac{\partial X}{\partial a} - \lambda X &= 0 \\ a^2 \frac{\partial^2 X}{\partial a^2} + a * \frac{\partial X}{\partial a} - a^2 \lambda X &= 0. \end{aligned} \quad (10)$$

Since $0 \leq x \leq L$, we also define this function over the interval $0 \leq a \leq 2\sqrt{L}$. We have arrived at the parametric form of the Bessel Equation of order zero. Its solutions are Bessel Functions of order zero.

Definition 4.6. Parametric Form of Bessel Functions of Order Zero and their Orthogonality

A Bessel Function of order zero is the solution to the Bessel Equation of order zero, which is a singular SLP. It is given by $y(x) = c_1 J_0(\mu x) + c_2 Y_0(\mu x)$. Since we are looking for bounded solutions, we can ignore the second term, given that $Y_0(x)$ is unbounded. Thus the bounded Bessel Function of order zero is given by:

$$y_j(x) = c_1 J_0(\mu_j x) = c_1 J_0(\sqrt{|\lambda|} x) \quad \text{where } J_0(\mu_j x) = \sum_{k=0}^{\infty} \frac{-1}{2^{2k} (k!)^2} x^{2k}.$$

The μ_j 's are given by $\frac{\alpha_j}{L}$ where α_j represents the j^{th} zero of the function and L is the length of the interval. Additionally, any distinct pair of two of these functions is orthogonal with respect to the weight function, $w(x) = 1$. For any one of these functions: $\langle J_0(\mu_j x), J_0(\mu_j x) \rangle_w = \int_0^a (J_0^2(\mu_j x) * x) dx = \frac{a^2}{2} * J_1^2(\alpha_j)$ where $J_1(x)$ is a Bessel Function of order 1.

For our function we see that $\mu_j = \frac{\alpha_j}{2\sqrt{L}}$. Thus we have $X_j(a) = c_1 J_0(\frac{\alpha_j}{2\sqrt{L}}a)$ and, plugging in $a = 2\sqrt{x}$, we find that:

$$X_j(x) = c_1 J_0(\alpha_j * \sqrt{\frac{x}{L}}) \text{ for } j = 1, 2, \dots \quad (11)$$

[1, Equation 6.3.8]. We also see that, using Definition 4.6, any distinct pair of these X_j 's is orthogonal with respect to the weight function $w(x) = 1$, after substitution. Additionally, for any one of these functions (using $a = 2\sqrt{x}$ and $da = \frac{1}{\sqrt{x}}dx$) their dot product is given by:

$$\begin{aligned} \langle J_0(\frac{\alpha_j}{2\sqrt{L}}a), J_0(\frac{\alpha_j}{2\sqrt{L}}a) \rangle_w &= \int_0^{2\sqrt{L}} (J_0^2(\frac{\alpha_j}{2\sqrt{L}}a) * a) da = 2 * \int_0^a J_0^2(\alpha_j * \sqrt{\frac{x}{L}}) dx \\ 2 * \langle J_0(\alpha_j * \sqrt{\frac{x}{L}}), J_0(\alpha_j * \sqrt{\frac{x}{L}}) \rangle_w &= \frac{(2\sqrt{L})^2}{2} J_1^2(\alpha_j) \\ \langle J_0(\alpha_j * \sqrt{\frac{x}{L}}), J_0(\alpha_j * \sqrt{\frac{x}{L}}) \rangle_w &= L * J_1^2(\alpha_j) \end{aligned} \quad (12)$$

We now plug our solutions for the equations for time and displacement, given by (6) and (11), respectfully, to the separated form, given by Theorem 4.1, $u(x, t) = X(x)T(t)$. We arrive at the following:

$$u_j(x, t) = J_0(\alpha_j * \sqrt{\frac{x}{L}}) (A_j \cos(\frac{\alpha_j}{2} \sqrt{\frac{g}{L}} t) + B_j \sin(\frac{\alpha_j}{2} \sqrt{\frac{g}{L}} t)), \quad \text{for } j = 1, 2, \dots$$

[1, Equation 6.3.9]. This gives us infinitely many eigenfunctions for the PDE described in Proposition 2.2. These eigenfunctions are orthogonal over an interval $0 \leq x \leq L$ with respect to the weight function $w(x) = 1$.

Theorem 4.7. *Eigenfunction Expansions* [1, Theorem 6.2.3]

If we have infinitely many eigenfunctions of a SLP that are all pairwise orthogonal over an interval with respect to a weight function, we can write:

$$f(x) = \sum_{j=1}^{\infty} c_j y_j(x), \quad \text{where } c_j = \frac{\langle f, y_j \rangle_w}{\langle y_j, y_j \rangle_w}.$$

Employing Theorem 4.7 we can find $u(x, t)$ by defining a summation over our u_j 's or, in other words, writing $u(x, t)$ as a linear combination of the u_j 's. Using this we arrive at the following:

Proposition 4.8. *General Solution to the PDE Defined in Proposition 2.2* [1, Equation 6.3.10]

$$u(x, t) = \sum_{j=1}^{\infty} J_0(\alpha_j * \sqrt{\frac{x}{L}}) (A_j \cos(\frac{\alpha_j}{2} \sqrt{\frac{g}{L}} t) + B_j \sin(\frac{\alpha_j}{2} \sqrt{\frac{g}{L}} t)).$$

Now we choose to impose the initial conditions, given in Proposition 3.1, to find the coefficients A_j and B_j . In Proposition 3.1 we have $u(x, 0) = f(x)$. Accordingly we see:

$$\begin{aligned} u(x, 0) = f(x) &= \sum_{j=1}^{\infty} J_0(\alpha_j * \sqrt{\frac{x}{L}}) (A_j \cos(\frac{\alpha_j}{2} \sqrt{\frac{g}{L}}(0)) + B_j \sin(\frac{\alpha_j}{2} \sqrt{\frac{g}{L}}(0))) \\ &= \sum_{j=1}^{\infty} A_j * J_0(\alpha_j * \sqrt{\frac{x}{L}}). \end{aligned} \quad (13)$$

We employ Theorem 4.7 and Equation 12 to find our coefficients A_j :

$$\begin{aligned} A_j &= \frac{\langle f(x), J_0(\alpha_j * \sqrt{\frac{x}{L}}) \rangle_w}{\langle J_0(\alpha_j * \sqrt{\frac{x}{L}}), J_0(\alpha_j * \sqrt{\frac{x}{L}}) \rangle_w} \\ &= \frac{1}{L * J_1^2(\alpha_j)} \int_0^L f(x) * J_0(\alpha_j * \sqrt{\frac{x}{L}}) dx \end{aligned} \quad (14)$$

[1, Equation 6.3.11]. To solve for the B_j 's we need to take the derivative of Proposition 4.8:

$$u_t(x, t) = \sum_{j=1}^{\infty} J_0(\alpha_j \sqrt{\frac{x}{L}}) (\frac{\alpha_j}{2} \sqrt{\frac{g}{L}}) (-A_j \sin(\frac{\alpha_j}{2} \sqrt{\frac{g}{L}} t) + B_j \cos(\frac{\alpha_j}{2} \sqrt{\frac{g}{L}} t)). \quad (15)$$

Imposing our initial condition $u_t(x, 0) = g(x)$, from Proposition 3.1, we have:

$$\begin{aligned} u_t(x, 0) = g(x) &= \sum_{j=1}^{\infty} J_0(\alpha_j \sqrt{\frac{x}{L}}) (\frac{\alpha_j}{2} \sqrt{\frac{g}{L}}) (-A_j \sin(\frac{\alpha_j}{2} \sqrt{\frac{g}{L}}(0)) + B_j \cos(\frac{\alpha_j}{2} \sqrt{\frac{g}{L}}(0))) \\ &= \sum_{j=1}^{\infty} B_j * J_0(\alpha_j \sqrt{\frac{x}{L}}) (\frac{\alpha_j}{2} \sqrt{\frac{g}{L}}). \end{aligned} \quad (16)$$

We will again apply Theorem 4.7 and Equation 12 to find the coefficient B_j 's:

$$\begin{aligned} B_j (\frac{\alpha_j}{2} \sqrt{\frac{g}{L}}) &= \frac{\langle g(x), J_0(\alpha_j * \sqrt{\frac{x}{L}}) \rangle_w}{\langle J_0(\alpha_j * \sqrt{\frac{x}{L}}), J_0(\alpha_j * \sqrt{\frac{x}{L}}) \rangle_w} \\ B_j &= \frac{2}{\sqrt{gL} * \alpha_j * J_1^2(\alpha_j)} \int_0^L g(x) * J_0(\alpha_j * \sqrt{\frac{x}{L}}) dx \end{aligned} \quad (17)$$

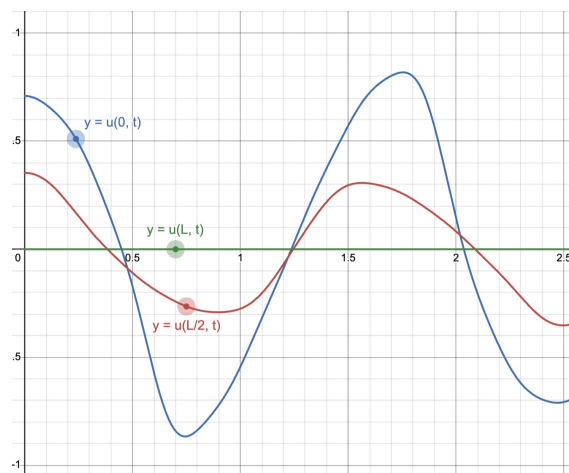
[1, Equation 6.3.12]. We have found the solution to the BVP described by Proposition 3.1. The solution is given by Proposition 4.8 where A_j and B_j are given by (13) and (16).

5. EXPLORATION OF RESULTS AND EXAMPLES

The calculations for this function, for different values of x and t , are very computationally heavy. To alleviate some of this workload we will use Desmos' calculator. To see these functions and change the parameters yourself you can use this [link](#) (it may take some time for graphs to load). Additionally, we approximate the Bessel Functions by using the first 20 terms. Finally, we approximate our solutions using the first 5 terms and the corresponding first 5 zeros of Bessel Functions of order 0. For the following examples we assume the chain has length $L = 1$. Additionally we observe a diagonal chain with no initial velocity. This means that we have the initial conditions: $f(x) = (1 - x) * \sin(\frac{\pi}{4}) = \frac{1-x}{\sqrt{2}}$ and $g(x) = 0$.

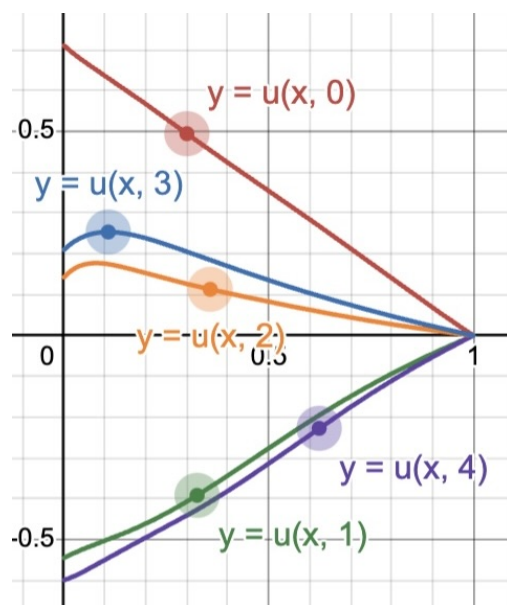
Example 5.1. Chain Motion at Different Points on Chain. Click [here](#) to see problem in Desmos.

Refer to the figure to the right. We first note that at $x = L$, for all time $t \geq 0$, the value of our displacement is 0. This is expected because, via our boundary conditions, we fixed this point on the chain in place, so we do not expect it to move. Additionally we observe that for $x = \frac{L}{2}$ and $x = 0$ their positions at $t = 0$ also reflect the boundary condition. We see that the chain oscillates back and forth around the equilibrium position ($u = 0$). The end of the chain ($x = 0$) oscillates with higher magnitude than the middle of the chain ($x = \frac{L}{2}$). This agrees with our intuition.



Example 5.2. Chain Motion at Different Times. Click [here](#) to see problem in Desmos.

Refer to the figure to the right. This figure shows the shape of the chain at 5 different times. We will note that at time $t = 0$, the chain takes on the shape of our initial condition (a diagonal chain), shown by the red line. At subsequent times we observe the motion of the chain. For all $x = L$ the position of the string is 0 which is reflective of our boundary condition (one end fixed). Additionally, as we progress from time $t = 1$ to $t = 4$ we can see a swinging motion in the chain. The chain oscillates around our equilibrium point $u = 0$. This agrees with our intuition.



6. CONCLUSION

The above paper derived and solved the Hanging Chain Problem, the physical analysis of a chain that is hanging vertically, fastened at one end, and then set into motion. The PDE for this equation describes the transverse oscillations of this chain. The solution to the above BVP, which followed from the PDE, gave us an equation to determine the horizontal displacement of the chain. We also explored the results of our BVP solution by analyzing some example physical situations. As a computer science major, I think it is important to consider how we might model physical situations which occur in everyday life. While it is often difficult to recreate these real-life occurrences accurately and descriptively, the content of Math 131P can very helpful for this endeavor. BVPs, and the techniques which are used in solving these problems, provide us with ways to approximate these situations. They give us a method through which we may represent physical environments via digital mediums and through which we may better understand and appreciate the world around us.

REFERENCES

- [1] N. H. Asmar. *Partial Differential Equations with Fourier Series and Boundary Value Problems*. 3rd ed. Dover Publications, 2017.